

Project Assignment 3: Exploring the Problem Space

Design Challenge

Redesign the online learning experience for first-generation university students in a world in which access to high-speed internet is unreliable

Task 1: Understand: What Do You Know About the Challenge?

Step A: Initial Mind Map

Things That Come to Mind When I Think About This Challenge

1. Connectivity: Unreliable internet means students can't access lectures, submit assignments, or participate in live sessions
2. First-Generation Status: These students lack family members who can guide them through university systems and expectations
3. Financial Constraints: Limited resources for devices, data plans, and backup connectivity options
4. Home Environment: Shared devices, competing family demands for bandwidth, lack of quiet study spaces
5. Isolation & Anxiety: Falling behind creates stress; missing out on social connections with peers

Chapter Headings for a Book About This Challenge

Chapter 1

The Digital Divide in Higher Education: Exploring how unreliable internet access creates inequities in learning outcomes

Chapter 2

Navigating the Unfamiliar: Understanding the unique challenges first-generation students face in higher education

Chapter 3

When Connectivity Fails: The lived experience of students during outages, lag, and limited bandwidth

Chapter 4

Beyond the Screen: Social, emotional, and psychological impacts of struggling to participate online

Chapter 5

Designing for Resilience: Creating learning experiences that work with, not against, connectivity constraints

Chapter 6

Voices from the Field: Stories and insights from students navigating these challenges

Chapter 7

A Path Forward: Recommendations for universities, educators, and policymakers

Key Issues and Opportunities by Chapter

Chapter 1: The Digital Divide

Issues: Unequal access to broadband; rural vs. urban disparities; affordability barriers

Opportunities: Low-bandwidth solutions; partnerships with telecom providers; mobile-first approaches

Chapter 2: Navigating the Unfamiliar

Issues: Lack of institutional knowledge; family cannot provide guidance; imposter syndrome

Opportunities: Peer mentoring programs; clear, simple communication; supportive onboarding

Chapter 3: When Connectivity Fails

Issues: Dropped connections during exams; inability to submit work; missing synchronous classes

Opportunities: Offline-first design; flexible deadlines; asynchronous alternatives

Chapter 4: Beyond the Screen

Issues: Isolation; mental health struggles; loss of campus community

Opportunities: Low-bandwidth social connection tools; hybrid community building; text-based discussion

Chapter 5: Designing for Resilience

Issues: One-size-fits-all solutions; assumptions about reliable connectivity

Opportunities: User-centered design; modular content; progressive enhancement

Relevant Users and Stakeholders

User/Stakeholder	Their Interest	Problems They Face
First-generation university students	Completing their degree successfully	Unreliable internet; lack of guidance; financial stress
University faculty	Delivering effective education; student success	Students not engaging; technical difficulties; extra workload
University IT departments	Maintaining systems; supporting users	Overwhelmed helpdesks; security concerns; legacy systems
University administrators	Retention; graduation rates; reputation	Meeting equity metrics; resource allocation; pressure to improve outcomes
Families of first-gen students	Supporting their children's success	Limited understanding of university systems; shared bandwidth; financial constraints
Employers	Hiring qualified graduates	Skills gaps; digital literacy concerns; equity in hiring

Step B: Deeper Exploration of Two Promising Chapters

Chapter 3: When Connectivity Fails

How has this aspect changed over time?

Online learning was once optional; now it's central. Students used to have campus computer labs and libraries as backup options. COVID-19 forced everything online, and many institutions have continued with hybrid or online models. The assumption of reliable connectivity has become normalized, making failure more consequential. Students who could once rely on campus Wi-Fi now face outages at home with fewer alternatives.

Are there specific solutions associated with this element?

- Downloadable materials
- Recorded lectures
- Email submission alternatives

How good or bad are these solutions?

These help but are often insufficient. Downloadable materials still require significant data. Recorded lectures don't allow for real-time questions. Email submissions assume students can access email reliably. None address the social or emotional impact of being absent from live sessions.

Open questions:

- How do we make learning engaging and social without requiring real-time participation?
- Can we design learning that actively works around connectivity issues rather than just coping with them?
- What low-bandwidth tools and approaches are we overlooking?

How I feel about this aspect

- *Concerned:* I worry about students feeling invisible and left behind
- *Curious:* I wonder what creative workarounds students have developed
- *Frustrated:* It feels unfair that access to education is limited by infrastructure beyond students' control

Chapter 4: Beyond the Screen

How has this aspect changed over time?

University has traditionally been about more than just academics, it's about relationships, networks, and belonging. Online learning has eroded this. Before, first-gen students already faced social challenges. Now they're doubly isolated: they don't have family guidance and they lack peer connections.

Are there specific solutions associated with this element?

- Discussion forums
- Virtual office hours
- Peer study groups

How good or bad are these solutions?

Discussion forums can feel performative and compulsory rather than genuinely social. Virtual office hours require scheduling and reliable connection. Study groups work but often exclude students who can't participate regularly. None address the fundamental need for belonging.

Open questions:

- How do we build genuine community when students can't rely on stable connectivity?

- What low-bandwidth social connection tools exist that we aren't using?
- How can we help first-gen students build networks they can draw on throughout their university journey?

How I feel about this aspect

- *Sad*: Students miss out on the social fabric of university life
- *Inspired*: There must be creative ways to build community that don't depend on high bandwidth
- *Empathetic*: I can imagine how lonely and disconnected this must feel

Three Most Interesting User Types and Their Problems

User Type	Their Problem
"The Rural Student", Lives in an area with limited or no broadband; relies on mobile data	Can't access video lectures; struggles to submit assignments; feels left out of class discussions
"The Sharing Student", Lives with family; shares one device and limited data	Competes with siblings for bandwidth; can't study at night when others need the internet; privacy concerns
"The Working Student", Works part-time or full-time; uses work Wi-Fi to study	Can only study in fits and starts; lacks a dedicated learning space; juggling work and academic pressures
"The Anxious Student" – Stressed about falling behind; feels invisible to professors and peers	Avoids reaching out for help; experiences imposter syndrome; disengages from learning
"The Resourceful Student" – Has developed creative workarounds (e.g., downloading at night, using library Wi-Fi)	Still falls behind; burnout from constant problem-solving; feels alone in their struggle

Task 2: Observing and Engaging Users

Step A: Questionnaire Design

Interview Guide (5 Open-Ended Questions)

Question 1: "Please tell me about the last time you had difficulty accessing your online classes or assignments because of internet problems.

- " *Purpose*: Understand concrete experiences of connectivity failure; capture emotional reactions

- *Probes:* "What happened next? How did you feel?"

Question 2: "What are the biggest challenges you face as a first-generation university student when it comes to online learning?"

- " *Purpose:* Surface issues beyond just internet access; understand broader student experience
- *Probes:* "What makes that challenging? Can you give me an example?"

Question 3: "Could you walk me through a typical day when you need to complete coursework online?"

- " *Purpose:* Understand the full context of the student experience; identify pain points
- *Probes:* "Where are you typically studying? What device are you using? Who else is sharing the connection?"

Question 4: "What would make online learning feel easier or more manageable for you?"

- " *Purpose:* Invite students to share ideas; surface unmet needs
- *Probes:* "What would that look like? Have you ever experienced anything like that?"

Question 5: "How do you stay connected to your professors and fellow students despite connectivity challenges?"

- " *Purpose:* Understand social and emotional aspects; identify existing workarounds
- *Probes:* "What works well for you? What doesn't work?"

Step B: Semi-Structured Interviews

Interview 1: "The Rural Student"

User Profile: Maria, 19, first-year university student, lives in a rural area with limited broadband access

Quotations:

"I have to drive 20 minutes to the public library just to watch my lecture videos. If I'm lucky, they'll have a quiet corner where I can study. But last week, the library closed early, and I had to leave in the middle of a lecture, I was so frustrated I almost cried."

"My professors keep saying 'just watch the recording later,' but they don't understand, downloading a 45-minute video uses up half my data allowance for the month."

"I've started screenshotting the slides and reading them like a textbook. It's not the same, I miss the explanations, but it's the only way I can keep up."

"I know I'm learning differently than my classmates. Will my degree mean the same thing when I graduate?"

"The hardest part is group projects. My group would meet on Zoom, but I couldn't join with video, and they'd forget I was there. I had to message them later to catch up on what I'd missed."

Emotional Context:

- Frustrated: Tired of the effort required just to participate
- Sad: Worries about falling behind
- Anxious: Questions whether her degree will hold the same value

Interview 2: "The Sharing Student"

User Profile: James, 20, second-year university student, lives with his family and shares a device and data plan

Quotations:

"There are five of us in the house, and we all share the same internet connection. My little brother is always watching YouTube or gaming, and my mom needs the Wi-Fi for her work. I end up studying after 11 pm, when everyone's asleep, but then I'm exhausted the next day."

"I've missed deadlines because I couldn't upload an assignment on time, the connection kept timing out. I have screenshots of the error messages to prove it wasn't my fault."

"My university has so many platforms. We use one for lectures, another for assignments, and a different one for discussion boards. Each of them loads differently when the connection is slow. It's a guessing game which one will work today."

"I feel like I'm always apologizing, to my professors for late work, to my group members for not being able to join meetings, to my family for using too much data."

"I want to ask for help, but I don't even know what to ask. I don't know what I don't know."

Emotional Context:

- Exhausted: Constantly negotiating with family
- Embarrassed: Always apologizing
- Overwhelmed: Navigating multiple platforms
- Isolated: Doesn't know how to ask for help

Interview 3: "The Working Student"

User Profile: Aisha, 22, third-year university student, works part-time and uses work Wi-Fi to study

Quotations:

"I work from 8 to 5, and my classes are in the evenings. The internet at my workplace is great, but I can't exactly sit there and attend a lecture while I'm supposed to be working. I try to download what I can during my lunch break and catch up after work, but honestly, I'm exhausted by then."

"Sometimes I attend lectures on my phone while I'm on my break. But I can't participate, and I'm always worried my manager will walk in and see me on my phone. It's stressful."

"I've never been to my professor's office hours. They're always during the day when I'm working. I tried emailing once, but it took three days to get a response, and by then, the assignment was due."

"My class has a WhatsApp group for students. It's the only way I can keep up with what's happening in the course. But I can only join the conversations at night, and by then, everything's already been discussed."

"I don't think my professors realize that some of us are working and studying at the same time. They think we're all just students with nothing else to do."

Emotional Context:

- Burned out: Juggling work and studies
- Resentful: Feels unseen by professors
- Overlooked: Believes her situation is ignored
- Determined: Keeps pushing despite challenges

Summary of Key Insights

Emerging Patterns from Interviews

1. Connectivity is just one part of the problem: Students face social, emotional, and logistical challenges that compound the technical issues
2. Students develop creative workaround: But these are exhausting and unsustainable
3. The "solutions" often don't understand the real constraints: Downloading recordings still uses data; office hours conflict with work schedules
4. Students feel invisible and unheard: They believe professors don't understand their situations
5. Social and community connections are fragile: Students miss out on relationships and peer learning

Surprising Findings

- One student used screenshots of slides as a substitute for watching lectures
- Group projects are particularly problematic for students with connectivity issues
- Students are often unaware of resources available to them (or don't know how to ask)

Open Questions from Interviews

- How can we design learning that works with students' constraints rather than requiring constant workarounds?
- How can we help students feel seen and supported without increasing their digital demands?
- What low-bandwidth alternatives to video lectures could be just as effective?